

Vertex Formula



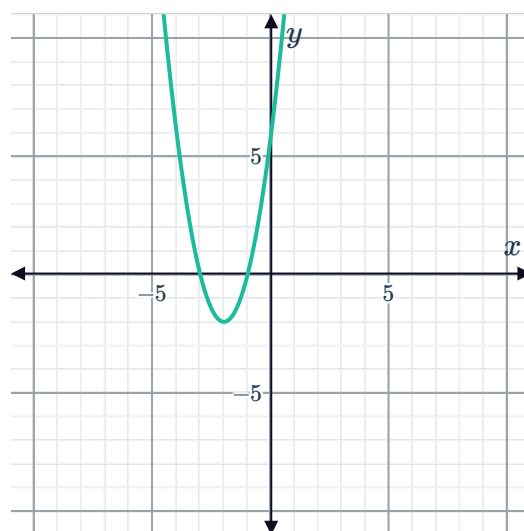
- 1 Consider the parabola of the form $y = ax^2 + bx + c$, where $a \neq 0$.
Complete the following statement:
The x -coordinate of the vertex of the parabola occurs at $x = \mathbb{I}$. The y -coordinate of the vertex is found by substituting $x = \mathbb{I}$ into the parabola's equation and evaluating the function at this value of x .
- 2 What does the line $x = \frac{-b}{2a}$ represent on the parabola defined by the equation $y = ax^2 + bx + c$ ($a \neq 0$)?
- 3 Consider the equation $y = 6x - x^2$.
 - a Find the x -intercepts of the quadratic function.
 - b Find the coordinates of the turning point.
- 4 Consider the parabola $y = x^2 - 18x + 10$.
 - a Is the graph of the function concave up or down?
 - b State the value of x at which the minimum value of the function occurs.
 - c State the minimum y -value of the function.
- 5 Consider the quadratic function $y = x^2 - 6x + 5$.
 - a Is the graph of the function concave up or down?
 - b Find the x -intercepts.
 - c State the y -intercept.
 - d Determine the axis of symmetry.
 - e Hence or otherwise state the coordinates of the vertex of the parabola.
- 6 Consider the quadratic function $y = -x^2 - 2x + 8$.
 - a Is the graph of the function concave up or down?
 - b Find the x -intercepts of the function.
 - c Find the y -intercept.
 - d Determine the axis of symmetry.
 - e Hence or otherwise find the vertex of the curve.

7 Consider the quadratic function $y = -8 - \text{[icon]} x^2$.

- a Is the graph of the function concave up or down?
- b Find the x -intercepts of the function.
- c Find the y -intercept.
- d Determine the axis of symmetry.
- e Hence or otherwise find the vertex of the curve.

8 Consider the function $y = x^2 + 4x + 3$.

- a Determine the equation of the axis of symmetry.
- b Hence determine the minimum value of y .
- c Hence state the coordinates of the vertex of the curve.
- d Here is the graph of another quadratic function. State the coordinates of its vertex.



- e Determine whether the following relationship is true between the graph of $y = x^2 + 4x + 3$ and the graph provided.
 - i They have the same x -intercepts.
 - ii They share the same turning point.
 - iii They have the same concavity.

9 Consider the function $y = (14 - x)(x - 6)$.

- a State the zeros of the function.
- b Find the axis of symmetry.
- c Is the graph of the function concave up or concave down?
- d Determine the maximum y -value of the function.

10 Find the coordinates of the vertex of $y = 3x^2 - 6x - 9$.

- 11 Find the maximum value of y for the quadratic function $y = -x^2 + 10x - 25$.
- 12 Consider the quadratic function $f(x) = -7.5x^2 - 1.3x - 1.3$.
- Find the x -coordinate of the vertex correct to three decimal places.
 - Hence, find the maximum value obtained by the function, correct to two decimal places.
- 13 Find the x -coordinate of the vertex of the parabola represented by $P(x) = px^2 - \frac{1}{2}px - q$.
- 14 Consider these two parabolas, labeled P_1 and P_2 .
- $$P_1 : y = x^2 + 4x + 6$$
- $$P_2 : y = x^2 - 4x + 6$$
- Determine the coordinates of the vertex for each parabola.
 - How far apart are the vertices of the two parabolas?
- 15 Consider the curve $y = x^2 + 6x + 4$.
- Determine the axis of symmetry.
 - Hence, determine the minimum value of y .

Applications

- 16 When an object is thrown into the air, its height above the ground is given by the equation
- $$h = 193 + 24s - s^2$$
- where s is its horizontal distance from where it was thrown.
- Find the value of s , at the point when it reaches its greatest height above the ground.
 - Find the maximum height reached by the object.
- 17 The height at time t of a ball thrown upwards is given by the equation $h = 59 + 42t - 7t^2$.
- How long does it take the ball to reach its maximum height?
 - Find the height of the ball at its highest point.